# Syllabus 9758

Statistics Summary

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# Contents


**1 Permutations and Combinations 2**

1.1 Counting Principles and Formulae . . . 2
1.2 Arrangements in a Line . . . 3
1.3 Circular Arrangements . . . 5
1.4 Selections and Groups . . . 6

**2 Probability 7**

2.1 Basic Rules . . . 7
2.2 Conditional Probability . . . 7
2.3 Independent and Mutually Exclusive Events . . . 8
2.4 Tree Diagrams . . . 9
2.5 Tables of Outcomes and Counting Methods . . . 9
2.6 Further Question Types . . . 10

**3 Discrete Random Variables 11**

3.1 Probability Distribution . . . 11
3.2 Expectation and Variance . . . 11
3.3 Linear Functions and Combinations . . . 12

**4 Binomial Distribution 13**

4.1 The Binomial Model . . . 13
4.2 Using the GC . . . 13
4.3 Common Question Types . . . 15

**5 Normal Distribution 17**

5.1 Properties . . . 17
5.2 Finding Probabilities and Values (GC) . . . 18
5.3 Linear Functions and Combinations of Normals . . . 19
5.4 Further Question Types . . . 19

**6 Sampling and the Central Limit Theorem 21**

6.1 Population and Sample . . . 21
6.2 Unbiased Estimates . . . 21
6.3 Distribution of the Sample Mean . . . 22
6.4 Which Distribution? . . . 22
6.5 Further Question Types . . . 23

**7 Hypothesis Testing 24**

7.1 Terminology . . . 24
7.2 Formulating the Hypotheses . . . 24
7.3 Test Statistic and Its Distribution . . . 24
7.4 The Test: $p$-value and Critical Region . . . 25
7.5 Using the Result of a Test to Find Unknowns . . . 26

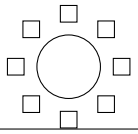
**8 Correlation and Linear Regression 27**

8.1 Scatter Diagrams . . . 27
8.2 Product Moment Correlation Coefficient $r$ . . . 27
8.3 Least-Squares Regression Line . . . 28
8.4 Interpolation, Extrapolation and Interpretation . . . 29
8.5 Transformations to Achieve Linearity . . . 29
8.6 Further Question Types . . . 30

# 1 Permutations and Combinations

## 1.1 Counting Principles and Formulae

| | Addition principle | Multiplication principle |
|---|---|---|
| Use when | either event $A$ **or** $B$ (cases) | stage 1 **and** stage 2 (steps) |
| Condition | cases cannot happen together | stages are done one after another |
| Total | $m+n$ | $m\times n$ |

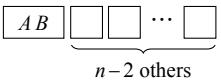
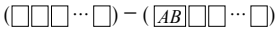
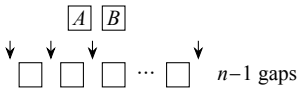
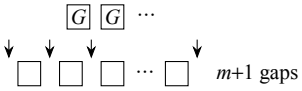
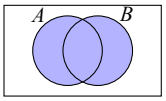
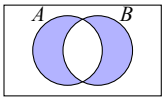
| Situation | Formula | Visual | Note |
|---|---|---|---|
| $n$ distinct objects in a row | $n!$ | $\boxed{n}\boxed{n-1}\boxed{n-2}\cdots\boxed{1}$ | $0!=1$ |
| $r$ of $n$ distinct objects, in order | ${}^nC_r\times r!=\dfrac{n!}{(n-r)!}$ | $\boxed{n}\boxed{n-1}\cdots\boxed{n-r+1}$ | order matters |
| $r$ of $n$ distinct objects, no order | ${}^nC_r=\dfrac{n!}{r!\,(n-r)!}$ | $\underbrace{\bullet\bullet\bullet\bullet}_{\text{pick } r}\bullet\bullet\bullet\cdots\bullet$ | ${}^nC_r={}^nC_{n-r}$ |
| $r$ places, each of $n$ choices, repeats allowed | $n^r$ | $\boxed{n}\boxed{n}\boxed{n}\cdots\boxed{n}$ | objects **can** be reused |
| $n$ objects, $p$ alike of one kind, $q$ alike of another | $\dfrac{n!}{p!\,q!}$ | $\underbrace{\boxed{a}\cdots\boxed{a}}_{p\text{ times}}\underbrace{\boxed{b}\cdots\boxed{b}}_{q\text{ times}}$ | divide by $k!$ for each $k$ identical |
| $n$ distinct objects in a circle | $\dfrac{n!}{n}=(n-1)!$ |  | rotations are the same |

**GC: nPr, nCr**: MATH → PRB. Type $n$ first, then the command, then $r$. ! is in the same menu.

**Note**

Order matters ⇒ select then arrange: ${}^nC_r\times r!$

# 1.2 Arrangements in a Line

| Condition | Method | Count | Visual |
|---|---|---|---|
| Fixed position (e.g. $A$ first; number is even; first digit $\neq 0$) | fill the restricted places **first**, then the rest | $\underline{k} \times _ \times \cdots$ |  |
| $A$, $B$ **together** | Grouping method: group $A$ and $B$ together, then arrange the blocks; arrange inside the block. | $(n-1)! \times 2!$ |   |
| $A$, $B$ **separated** (complement) | total − together | $n! - (n-1)!\,2!$ |  |
| $A$, $B$ **separated** (slot in) | slot in: arrange the other $n-2$; choose 2 of the $n-1$ gaps for $A$, $B$; arrange $A$, $B$ | $(n-2)! \times {}^{n-1}\mathrm{C}_2 \times 2!$ |   |
| No two of a group adjacent | **slot in method**: arrange the other $m$; choose $k$ of the $m+1$ gaps for the group; arrange the group | $m! \times {}^{m+1}\mathrm{C}_k \times k!$ |   |
| "at least one" | complement: total − none | | |
| $A$ **or** $B$ (or both) | add, then subtract the overlap once | $n(A)+n(B)-n(A\cap B)$ |   |
| $A$ **or** $B$ but not both | add, then subtract the overlap twice | $n(A)+n(B)-2n(A\cap B)$ |   |
| Several cases (e.g. last digit 0 or not) | case table: count each case, then **add** | | |

**Example 1.1**

Find the number of arrangements of the letters in the word STATISTICS.

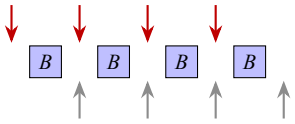
$$\text{S}\times 3,\ \text{T}\times 3,\ \text{I}\times 2$$

$$\text{total} = \frac{10!}{3!\,3!\,2!}$$

$$= 50\,400$$

**Example 1.2**

Find the number of ways 4 boys and 4 girls can be arranged in a line such that boys and girls alternate.

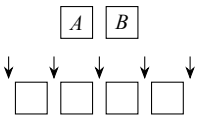



$$\text{total} = 2\times 4!\times 4!$$

$$= 1152$$

**Example 1.3**

How many ways are there to arrange 6 people in a line with $A$ and $B$ separated?

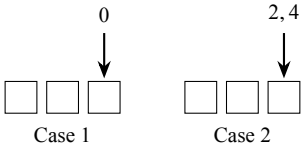



$$\text{total} = 4!\times {}^5\text{C}_2\times 2!$$

$$= 480$$

**Example 1.4**

Even 3-digit numbers from 0, 1, 2, 3, 4, 5, no repeats



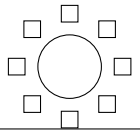
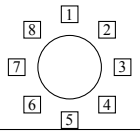
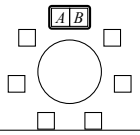
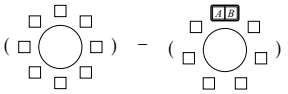
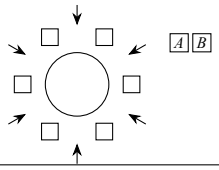
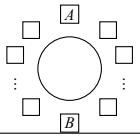
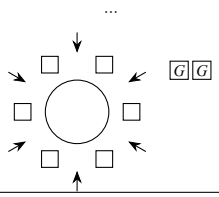

$$\text{last } 0 = 5\times 4 = 20$$

$$\text{last } 2 \text{ or } 4 = 2\times 4\times 4 = 32$$

$$\therefore \text{total} = 20+32$$

$$= 52$$

# 1.3 Circular Arrangements

| Situation | Method | Count | Visual |
|---|---|---|---|
| $n$ people round a table | | $(n-1)!$ |  |
| seats **numbered / distinguishable** | ignore the numbered seats and do everything as per normal, then multiply by the number of seats at the end | $(n-1)!\times n$ |  |
| $A$, $B$ **together** | grouping, then circular arrangement of the blocks | $(n-2)!\times 2!$ |  |
| $A$, $B$ **separated** (complement) | total − together | $(n-1)!-(n-2)!\times 2!$ |  |
| $A$, $B$ **separated** (slot in) | slot in: arrange the other $n-2$ in a circle; choose 2 of the $n-2$ gaps for $A$, $B$; arrange $A$, $B$ | $(n-3)!\times {}^{n-2}\mathrm{C}_2\times 2!$ |  |
| $A$, $B$ in fixed relation (e.g. $A$ opposite $B$) | fix $A$; $B$ has 1 place; arrange the rest | $(n-2)!$ |  |
| no two of $k$ people adjacent | **slot in method**: arrange the other $m$ in a circle; choose $k$ of the $m$ gaps; arrange the $k$ people | $(m-1)!\times {}^{m}\mathrm{C}_k\times k!$ |  |

## 1.4 Selections and Groups

| Situation | Method | Count |
|---|---|---|
| choose $r$ of $n$, with a condition (e.g. $\geq 1$ woman) | ${}^nC_r$; cases (add), or complement | total − none |
| choose, **then** arrange | select, then multiply by $r!$ (or by the arrangements) | ${}^nC_r \times r!$ |
| "at least one of **each** kind" | consider cases<br>Do **not** choose one of each first, then mix the rest together and choose the remaining (overcounts) | add cases |
| forming words / codes with **repeated** letters | cases by the pattern of repeats (all different; one pair; …): choose the letters, then arrange $\frac{r!}{p!\,q!}$ | add the cases |
| groups of **different** sizes / labelled groups | choose group by group; named teams, different tasks or sizes: **no** division | ${}^nC_a \, {}^{n-a}C_b \cdots$ |
| $k$ groups of the **same** size, unlabelled | as above, then $\div k!$ (identical groups of equal size) | $\frac{{}^nC_a \, {}^{n-a}C_a \cdots}{k!}$ |

**Example 1.5**

8 people into two groups of 4

$$\text{unlabelled} = \frac{{}^8C_4}{2!} = 35$$

$$\text{teams } X, Y = {}^8C_4 = 70$$

**Example 1.6**

3-letter codes from MISSISSIPPI

$$\text{all different} = {}^4C_3 \times 3! = 24$$

$$\text{one pair} = (3 \times 3) \times \frac{3!}{2!} = 27$$

$$\text{all alike} = 2$$

$$\therefore \text{total} = 53$$

**Example 1.7**

5 from 3 red, 4 green, 5 blue; every colour used

$$(3, 1, 1) = {}^3C_3 \times {}^4C_1 \times {}^5C_1 = 20$$

$$(1, 3, 1) = {}^3C_1 \times {}^4C_3 \times {}^5C_1 = 60$$

$$(1, 1, 3) = {}^3C_1 \times {}^4C_1 \times {}^5C_3 = 120$$

$$(2, 2, 1) = {}^3C_2 \times {}^4C_2 \times {}^5C_1 = 90$$

$$(2, 1, 2) = {}^3C_2 \times {}^4C_1 \times {}^5C_2 = 120$$

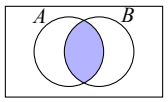
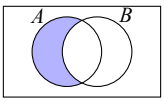
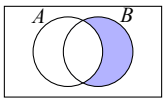
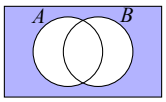
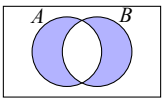
$$(1, 2, 2) = {}^3C_1 \times {}^4C_2 \times {}^5C_2 = 180$$

$$\therefore \text{ways} = 20 + 60 + 120 + 90 + 120 + 180 = 590$$

# 2 Probability

## 2.1 Basic Rules

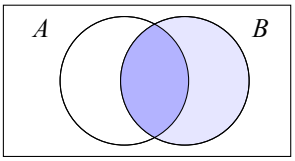
| Result | Formula | Note |
|---|---|---|
| Equally likely outcomes | $\mathrm{P}(A)=\frac{n(A)}{n(S)}$ | $0\leqslant \mathrm{P}(A)\leqslant 1$ |
| Complement | $\mathrm{P}(A')=1-\mathrm{P}(A)$ | "at least one" = 1 − none |
| Addition rule | $\mathrm{P}(A\cup B)=\mathrm{P}(A)+\mathrm{P}(B)-\mathrm{P}(A\cap B)$ | $A\cup B$: $A$ or $B$ (or both) |
| Mutually exclusive | $\mathrm{P}(A\cap B)=0$ | so $\mathrm{P}(A\cup B)=\mathrm{P}(A)+\mathrm{P}(B)$ |
| Exhaustive | $\mathrm{P}(A\cup B)=1$ | |

|  $A\cap B$ |  $A\cap B'$ |  $A'\cap B$ |  $A'\cap B'=(A\cup B)'$ |  exactly one |
|---|---|---|---|---|
| $\mathrm{P}(A\cap B)$ | $\mathrm{P}(A)-\mathrm{P}(A\cap B)$ | $\mathrm{P}(B)-\mathrm{P}(A\cap B)$ | $1-\mathrm{P}(A\cup B)$ | $\mathrm{P}(A)+\mathrm{P}(B)-2\mathrm{P}(A\cap B)$ |

The four regions add up to 1. $(A\cup B)'=A'\cap B'$ and $(A\cap B)'=A'\cup B'$. None of these probability results is in MF27.

## 2.2 Conditional Probability

| Result | Formula | Note |
|---|---|---|
| Conditional probability | $\mathrm{P}(A\mid B)=\frac{\mathrm{P}(A\cap B)}{\mathrm{P}(B)}$ | probability of $A$ given $B$ has happened; $\mathrm{P}(B)\neq 0$ |
| Multiplication rule | $\mathrm{P}(A\cap B)=\mathrm{P}(A\mid B)\mathrm{P}(B)=\mathrm{P}(B\mid A)\mathrm{P}(A)$ | used along a tree branch |
| Complement given $B$ | $\mathrm{P}(A'\mid B)=1-\mathrm{P}(A\mid B)$ | |
| Warning | $\mathrm{P}(A\mid B)\neq \mathrm{P}(B\mid A)$ | different denominators |

reduced sample space: 


$$\mathrm{P}(A\mid B)=\frac{\text{shaded}}{\text{whole circle } B}$$

## 2.3 Independent and Mutually Exclusive Events

| | Independent | Mutually exclusive |
|---|---|---|
| Meaning | one event does not change the probability of the other | cannot both happen |
| Test | • $\mathrm{P}(A\cap B)=\mathrm{P}(A)\mathrm{P}(B)$<br>• $\mathrm{P}(A\mid B)=\mathrm{P}(A)$<br>• $\mathrm{P}(B\mid A)=\mathrm{P}(B)$ | • $\mathrm{P}(A\cap B)=0$<br>• $\mathrm{P}(A\cup B)=\mathrm{P}(A)+\mathrm{P}(B)$ |
| Venn | circles overlap, $\mathrm{P}(A\cap B)=\mathrm{P}(A)\mathrm{P}(B)$ | circles do not overlap |

**Note**

- If $A$, $B$ are independent, so are $A$ and $B'$, $A'$ and $B$, $A'$ and $B'$.
- If $\mathrm{P}(A)>0$ and $\mathrm{P}(B)>0$, events cannot be both independent and mutually exclusive: $\mathrm{P}(A\cap B)=0\neq\mathrm{P}(A)\mathrm{P}(B)$.
- To show independence, show $\mathrm{P}(A\cap B)=\mathrm{P}(A)\mathrm{P}(B)$; *do not* assume it from the wording.
- Independence cannot be detected from the shape of the Venn diagram: it generally looks the same as a Venn diagram in which $A$ and $B$ are not independent.

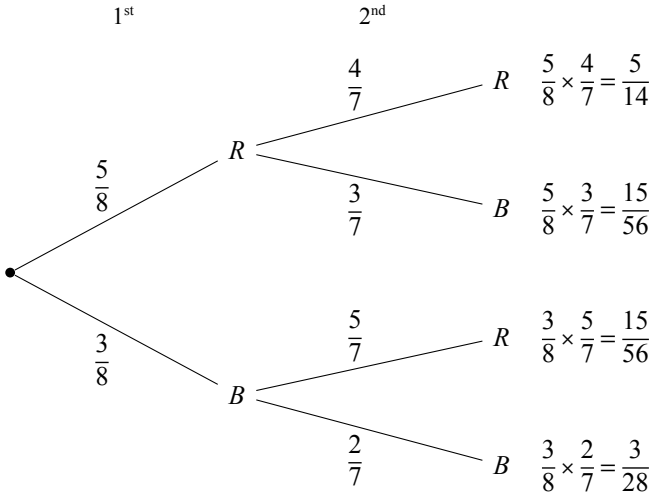
**Example 2.1**

Given that $A$ is independent of $B$, prove that $A'$ is independent of $B'$.

$$\begin{aligned}\mathrm{P}(A'\cap B') &= 1-\mathrm{P}(A\cup B)\\ &= 1-\mathrm{P}(A)-\mathrm{P}(B)+\mathrm{P}(A\cap B)\\ \because\ \mathrm{P}(A\cap B) &= \mathrm{P}(A)\mathrm{P}(B)\\ \therefore\ \mathrm{P}(A'\cap B') &= 1-\mathrm{P}(A)-\mathrm{P}(B)+\mathrm{P}(A)\mathrm{P}(B)\\ &= (1-\mathrm{P}(A))(1-\mathrm{P}(B))\\ &= \mathrm{P}(A')\mathrm{P}(B')\\ \therefore\ A',\ B' &\text{ are independent}\end{aligned}$$

## 2.4 Tree Diagrams

Multiply along branches (∩); add across leaves (∪). Branches from one node sum to 1. Without replacement: probabilities in the second stage **change**.




Bag: 5 red, 3 blue; 2 drawn without replacement.

$$\begin{aligned}\mathrm{P}(\text{different colours}) &= \frac{5}{8}\times\frac{3}{7}+\frac{3}{8}\times\frac{5}{7}\\ &= \frac{15}{56}+\frac{15}{56}\\ &= \frac{15}{28}\end{aligned}$$

$$\begin{aligned}\mathrm{P}(1\text{st } R \mid 2\text{nd } B) &= \frac{\mathrm{P}(1\text{st } R \text{ and } 2\text{nd } B)}{\mathrm{P}(2\text{nd } B)}\\ &= \frac{\mathrm{P}(R,B)}{\mathrm{P}(R,B \text{ or } B,B)}\\ &= \frac{\frac{15}{56}}{\frac{15}{56}+\frac{6}{56}}\\ &= \frac{5}{7}\end{aligned}$$

## 2.5 Tables of Outcomes and Counting Methods

| + | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| 5 | 6 | 7 | 8 | 9 | 10 | 11 |
| 6 | 7 | 8 | 9 | 10 | 11 | 12 |

two dice: $n(S)=36$

$$\mathrm{P}(\text{sum}=7)=\frac{6}{36}=\frac{1}{6}$$

$$\mathrm{P}(\text{sum}\geqslant 10)=\frac{6}{36}=\frac{1}{6}$$

$$\mathrm{P}(\text{sum}=7 \mid \text{a 3 shown})=\frac{2}{11}$$

**Example 2.2**

There are 6 men and 10 women. 5 people are chosen. Find the probability that the group has 3 men.

(a)

$$\begin{aligned}\mathrm{P}(3M, 2W) &= \frac{6}{16}\times\frac{5}{15}\times\frac{4}{14}\times\frac{10}{13}\times\frac{9}{12}\times\frac{5!}{2!\,3!}\\ &= \frac{15}{728}\times 10\\ &= \frac{75}{364}\end{aligned}$$

(b)

$$\begin{aligned}\mathrm{P}(3 \text{ men}) &= \frac{{}^{6}\mathrm{C}_3\times{}^{10}\mathrm{C}_2}{{}^{16}\mathrm{C}_5}\\ &= \frac{20\times 45}{4368}\\ &= \frac{75}{364}\end{aligned}$$

## 2.6 Further Question Types

| Type | Method |
|---|---|
| Range of $\mathrm{P}(A\cap B)$ | Label $\mathrm{P}(A\cap B)=x$, then label all other areas in terms of $x$. Let each area $\geqslant 0$ and solve. |
| Games: "first to win", taking turns | list the winning sequences; the game can go on forever; the probabilities form a **GP**; $S_\infty = \frac{a}{1-r}$ $(\lvert r\rvert<1)$ |
| Venn diagram with unknowns ($x$, $y$) | label the regions (2 or 3 events) with $x$, $y$, …; regions add to 1; use some other info in the equation to link $x$ and $y$; every region $\geqslant 0$ gives the greatest / least value |

**Example 2.3**

$\mathrm{P}(A)=0.45$, $\mathrm{P}(B)=0.7$. Find the range of values of $\mathrm{P}(A\cap B)$.

let $\mathrm{P}(A\cap B)=x$

$\mathrm{P}(A\cap B')=0.45-x$

$\mathrm{P}(A'\cap B)=0.7-x$

$\mathrm{P}(A'\cap B')=1-(0.45-x)-(0.7-x)-x$

$=x-0.15$

$\because$ each area $\geqslant 0$

$\therefore 0.15\leqslant \mathrm{P}(A\cap B)\leqslant 0.45$

**Example 2.4**

$A$, $B$ take turns to roll a die, $A$ first; first six wins

$$\begin{aligned}\mathrm{P}(A \text{ wins}) &= \frac{1}{6}+\left(\frac{5}{6}\right)^2\frac{1}{6}+\left(\frac{5}{6}\right)^4\frac{1}{6}+\cdots\\ &= \frac{1/6}{1-25/36}\\ &= \frac{6}{11}\end{aligned}$$

# 3 Discrete Random Variables

## 3.1 Probability Distribution

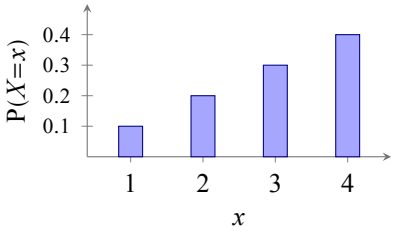
| Item | Result | Note |
|---|---|---|
| Discrete r.v. $X$ | takes separate values $x_1, x_2, \ldots$ (counts) | e.g. number of heads |
| Probability distribution | table of $x$ against $\mathrm{P}(X=x)$ | |
| Conditions | $0 \leqslant \mathrm{P}(X=x) \leqslant 1$ and $\sum_{\text{all } x} \mathrm{P}(X=x) = 1$ | use to find an unknown constant |

| $x$ | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| $\mathrm{P}(X=x)$ | $k$ | $2k$ | $3k$ | $4k$ |

$$\sum \mathrm{P}(X=x) = 1$$

$$10k = 1$$

$$\therefore k = 0.1$$




## 3.2 Expectation and Variance

| Result | Formula | Note |
|---|---|---|
| Expectation (mean) | $\mathrm{E}(X) = \mu = \sum x\,\mathrm{P}(X=x)$ | long-run average |
| Expectation of a function | $\mathrm{E}(g(X)) = \sum g(x)\,\mathrm{P}(X=x)$ | e.g. $\mathrm{E}(X^2) = \sum x^2 \mathrm{P}(X=x)$ |
| Variance | $\mathrm{Var}(X) = \mathrm{E}[(X-\mu)^2] = \mathrm{E}(X^2) - [\mathrm{E}(X)]^2$ | $\mathrm{Var}(X) \geqslant 0$ |
| Standard deviation | $\sigma = \sqrt{\mathrm{Var}(X)}$ | |

**GC:** enter $x$ in **L1**, $\mathrm{P}(X=x)$ in **L2**. **STAT** → **CALC** → **1-Var Stats L1,L2**. Then $\bar{x} = \mathrm{E}(X)$, $\sigma\mathbf{x} = \sigma$ (not **Sx**); $\mathrm{Var}(X) = \sigma^2$. $\mathrm{E}(X^2) = \mathrm{Var}(X) + [\mathrm{E}(X)]^2$, or read $\Sigma\mathbf{x}^2$.

$$\mathrm{E}(X) = 1(0.1) + 2(0.2) + 3(0.3) + 4(0.4) = 3$$

$$\mathrm{E}(X^2) = 1(0.1) + 4(0.2) + 9(0.3) + 16(0.4) = 10$$

$$\mathrm{Var}(X) = 10 - 3^2 = 1$$

## 3.3 Linear Functions and Combinations

| Result | Formula | Condition |
|---|---|---|
| Linear function | $\mathrm{E}(aX+b)=a\mathrm{E}(X)+b$ | always |
| | $\mathrm{Var}(aX+b)=a^2\mathrm{Var}(X)$ | always; $b$ does not change spread |
| Sum / difference | $\mathrm{E}(aX\pm bY)=a\mathrm{E}(X)\pm b\mathrm{E}(Y)$ | always |
| | $\mathrm{Var}(aX\pm bY)=a^2\mathrm{Var}(X)+b^2\mathrm{Var}(Y)$ | $X$, $Y$ **independent**; variances **add** for $\pm$ |
| Sum of $n$ observations | $\mathrm{E}(X_1+X_2+\cdots+X_n)=n\mu$ | always (independence not needed) |
| | $\mathrm{Var}(X_1+X_2+\cdots+X_n)=n\sigma^2$ | $X_1,\ldots,X_n$ **independent** observations of $X$ |
| One observation $\times n$ | $\mathrm{E}(nX)=n\mu$ | always |
| | $\mathrm{Var}(nX)=n^2\sigma^2$ | always; $\mathrm{Var}(nX)\neq\mathrm{Var}(X_1+\cdots+X_n)$ |

**Example 3.1**

$Y=3X-2$, using $\mathrm{E}(X)=3$, $\mathrm{Var}(X)=1$

$$\mathrm{E}(Y)=3(3)-2=7$$

$$\mathrm{Var}(Y)=3^2(1)=9$$

**Example 3.2**

Prize $W$: 0, 4, 10 with probabilities 0.6, 0.3, 0.1; cost to play \$3

$$\mathrm{E}(W)=0+1.2+1=2.2$$

$$\mathrm{E}(\text{gain})=2.2-3=-0.8$$

$\therefore$ not a fair game

**Note**

- If a game is **fair**, $\mathrm{E}(\text{gain})=0$.
- To find the expected value of a variable other than the original $X$, there are two options:
  (1) link the new variable to the old one, e.g. $Y=5-2X$, then apply the expectation formulas;
  (2) draw an entirely new table for the new variable and find the expectation directly from that table.

# 4 Binomial Distribution

## 4.1 The Binomial Model

| Item | Result | Note |
|---|---|---|
| Notation | $X \sim \mathrm{B}(n,p)$ | $X$ = no. of successes in $n$ trials |
| Probability | $\mathrm{P}(X=r) = {}^{n}\mathrm{C}_{r}\, p^{r}(1-p)^{n-r}, \quad r=0,1,\ldots,n$ | MF27 |
| Mean | $\mathrm{E}(X) = np$ | MF27 (no proof) |
| Variance | $\mathrm{Var}(X) = np(1-p)$ | MF27 (no proof) |

**Conditions for $X \sim \mathrm{B}(n,p)$ (state in context)**

- fixed number $n$ of trials
- each trial has exactly 2 outcomes: success / failure
- $\mathrm{P}(\text{success}) = p$ is the **same** for every trial
- trials are **independent**

Note

$X \sim \mathrm{B}(n,p)$ counts the number of successes out of $n$ trials. So $0 \leqslant X \leqslant n$, $X \in \mathbb{Z}$.

| Model fails when … | Why |
|---|---|
| selecting without replacement from a small population | $p$ changes; trials not independent |
| counting trials until the first success | $n$ is not fixed |
| events are linked (e.g. family members, same day) | trials not independent |
| $p$ varies (e.g. by time of day, by type of item) | $p$ not constant |

## 4.2 Using the GC

| Probability | Conversion | GC command |
|---|---|---|
| $\mathrm{P}(X=r)$ | not needed | `binompdf(n,p,r)` |
| $\mathrm{P}(X \leqslant r)$ | not needed | `binomcdf(n,p,r)` |
| $\mathrm{P}(X<r)$ | $\mathrm{P}(X \leqslant r-1)$ | `binomcdf(n,p,r-1)` |
| $\mathrm{P}(X \geqslant r)$ | $1-\mathrm{P}(X \leqslant r-1)$ | `1-binomcdf(n,p,r-1)` |
| $\mathrm{P}(X>r)$ | $1-\mathrm{P}(X \leqslant r)$ | `1-binomcdf(n,p,r)` |
| $\mathrm{P}(a \leqslant X \leqslant b)$ | $\mathrm{P}(X \leqslant b)-\mathrm{P}(X \leqslant a-1)$ | `binomcdf(n,p,b)-binomcdf(n,p,a-1)` |

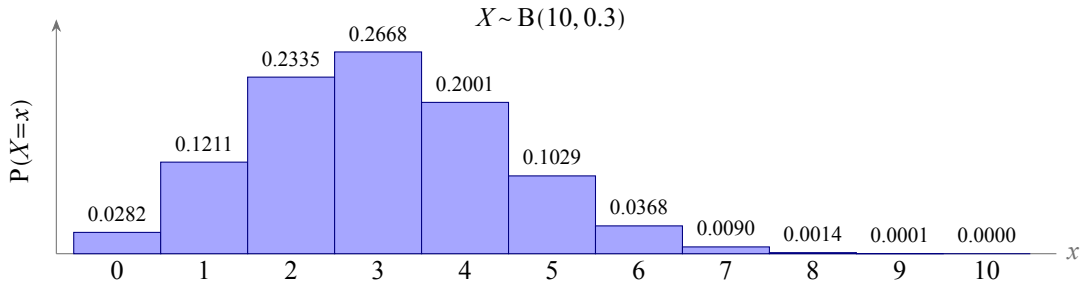
**Note**

- **binompdf(…)**: $P(X=r)$
- **binomcdf(…)**: $P(X \leqslant r)$

Any other inequalities must be converted to the above two forms before keying them into your GC.

**GC:** 2nd VARS (DISTR) → **binompdf(** / **binomcdf(**.

**GC:** solve for $p$ or $n$: Y=, $Y_1 = $ **binomcdf(10,X,3)** (X plays $p$), $Y_2 =$ the given value; GRAPH; 2nd TRACE (CALC) → **5:intersect**. For $n$: use 2nd GRAPH (TABLE) instead.




**Example 4.1**

$X \sim \mathrm{B}(10, 0.3)$

$$\begin{aligned} \mathrm{P}(X=3) &= {}^{10}\mathrm{C}_3 (0.3)^3 (0.7)^7 \\ &= 0.267 \\ \mathrm{P}(X \geqslant 4) &= 1 - \mathrm{P}(X \leqslant 3) \\ &= 1 - 0.6496 = 0.350 \end{aligned}$$

$$\mathrm{E}(X) = 3, \ \mathrm{Var}(X) = 2.1$$

**Example 4.2**

$X \sim \mathrm{B}(20, 0.4)$. Find the probability that $X$ is within 2 standard deviations of the mean.

$$\begin{aligned} \mu &= 20(0.4) = 8 \\ \sigma &= \sqrt{20(0.4)(0.6)} \\ &= \sqrt{4.8} = 2.1909 \\ \mu - 2\sigma &= 3.618 \\ \mu + 2\sigma &= 12.382 \\ \mathrm{P}(3.618 < X < 12.382) &= \mathrm{P}(4 \leqslant X \leqslant 12) \\ &= \mathrm{P}(X \leqslant 12) - \mathrm{P}(X \leqslant 3) \\ &= 0.97897 - 0.01596 \\ &= 0.963 \end{aligned}$$

**Example 4.3**

$X \sim \mathrm{B}(10, 0.3)$. Find the probability that the tenth result is the 3rd success.

$$\text{let } Y \sim \mathrm{B}(9, 0.3)$$

$$\mathrm{P}(Y=2) = {}^{9}\mathrm{C}_2 (0.3)^2 (0.7)^7$$

$$= 0.26683$$

$$\therefore \mathrm{P}(\text{10th is 3rd success}) = \mathrm{P}(Y=2) \times 0.3$$

$$= 0.26683 \times 0.3$$

$$= 0.0800$$

## 4.3 Common Question Types

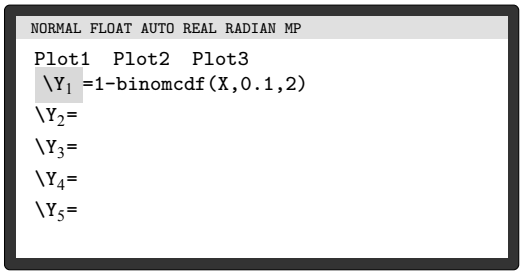
| Type | Method |
|---|---|
| Smallest $n$ with $\mathrm{P}(X \geqslant 1) > c$ | Let $X \sim \mathrm{B}(n, p)$ ($p$ is given), then use **GC table** to solve for $n$. |
| Unknown $p$ | form the equation in $p$ (e.g. $\mathrm{P}(X=0) = (1-p)^n$); solve; or use GC graph to solve for $p$ |
| Given a condition | conditional probability: $\mathrm{P}(X=r \mid X \geqslant 1) = \dfrac{\mathrm{P}(X=r)}{\mathrm{P}(X \geqslant 1)}$, $r \geqslant 1$ |
| Most probable value (mode) $r$ | **Compare mode $r$ with its directly adjacent neighbours**<br>• $\mathrm{P}(X=r) > \mathrm{P}(X=r+1)$<br>• $\mathrm{P}(X=r) > \mathrm{P}(X=r-1)$<br>Equivalently, solve $\dfrac{\mathrm{P}(X=r)}{\mathrm{P}(X=r+1)} > 1$ |
| Trial $r$ is the $k$-th success | first $r-1$ trials contain $k-1$ successes, then a success: ${}^{r-1}\mathrm{C}_{k-1} p^{k-1} (1-p)^{r-k} \times p$<br>Alternatively, $\underbrace{{}^{r-1}\mathrm{C}_{k-1} p^{k-1} (1-p)^{r-k}}_{\text{define } Y \sim \mathrm{B}(r-1, p), \text{ and } \mathrm{P}(Y=k-1)} \times p$ |
| Total / mean of $N$ independent copies of $X \sim \mathrm{B}(n, p)$ ($N$ is large) | Use the CLT (Ch 6) with $\mu = np$, $\sigma^2 = np(1-p)$<br>• $\bar{X} \sim \mathrm{N}\left(\mu, \dfrac{\sigma^2}{N}\right) = \mathrm{N}\left(np, \dfrac{np(1-p)}{N}\right)$ approximately<br>• $X_1 + X_2 + \cdots + X_N \sim \mathrm{N}(N\mu, N\sigma^2) = \mathrm{N}(Nnp, Nnp(1-p))$ approximately<br>For use of CLT, usually coupled with the phrases "estimate the probability" or "by using a suitable approximation". |
| Estimate $p$ from data | $p \approx \dfrac{\text{total successes}}{\text{total trials}} = \dfrac{\bar{x}}{n}$ (frequency table: $\sum xf / n \sum f$) |
| Sketch of $\mathrm{B}(n, p)$ | vertical bars at $r = 0, 1, \ldots, n$; label each probability above the bars; skewed right if $p < 0.5$, left if $p > 0.5$, symmetric if $p = 0.5$ |
| Double binomial | new $p'$ from the first part, e.g. $Y \sim \mathrm{B}(m, p')$ where $p' = \mathrm{P}(X \geqslant 2)$ ("on how many of $m$ days") |

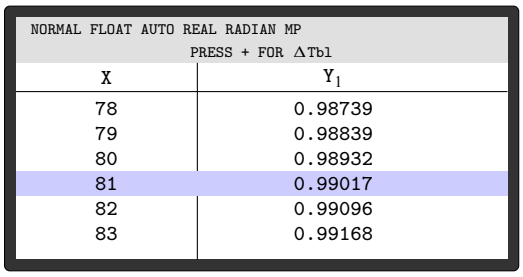
**Example 4.4**

$X \sim \mathrm{B}(n, 0.1)$. Find the smallest value of $n$ such that $\mathrm{P}(X \geqslant 3) > 0.99$.

$$\mathrm{P}(X \geqslant 3) > 0.99$$

$$1 - \mathrm{P}(X \leqslant 2) > 0.99$$




$$\therefore n_{\min} = 81$$

**Example 4.5**

$X \sim \mathrm{B}(10, p)$. Given that the mode of $X$ is 3, find the range of $p$.

$$\mathrm{P}(X=3) > \mathrm{P}(X=2)$$

$${}^{10}\mathrm{C}_3 p^3 (1-p)^7 > {}^{10}\mathrm{C}_2 p^2 (1-p)^8$$

$$120p > 45(1-p)$$

$$165p > 45$$

$$p > \frac{3}{11}$$

$$\mathrm{P}(X=3) > \mathrm{P}(X=4)$$

$${}^{10}\mathrm{C}_3 p^3 (1-p)^7 > {}^{10}\mathrm{C}_4 p^4 (1-p)^6$$

$$120(1-p) > 210p$$

$$120 > 330p$$

$$p < \frac{4}{11}$$

$$\therefore \frac{3}{11} < p < \frac{4}{11}$$

**Example 4.6**

The number of defective items in a box of 20 is modelled by $X \sim \mathrm{B}(20, 0.3)$. A random sample of 50 boxes is taken. By using a suitable approximation, estimate the probability that the mean number of defective items per box exceeds 6.2.

$$\mathrm{E}(X) = 20(0.3) = 6$$

$$\mathrm{Var}(X) = 20(0.3)(0.7) = 4.2$$

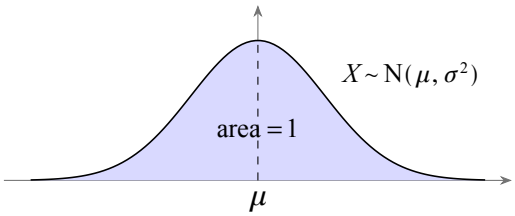
$\because N = 50$ is large, by the Central Limit Theorem,

$$\bar{X} \sim \mathrm{N}\left(6, \frac{4.2}{50}\right) \text{ approximately}$$

$$\therefore \mathrm{P}(\bar{X} > 6.2) = 0.245$$

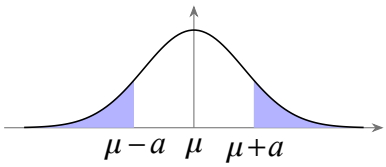
# 5 Normal Distribution

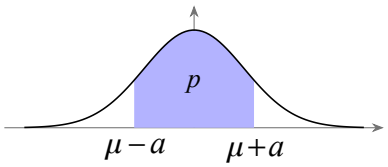
## 5.1 Properties




- bell-shaped, symmetric about $x = \mu$
- total area = 1; mean = $\mu$; variance = $\sigma^2$
- continuous: $\mathrm{P}(X = a) = 0$, so $\mathrm{P}(X < a) = \mathrm{P}(X \leqslant a)$
- $\sigma$ larger ⇒ wider and flatter

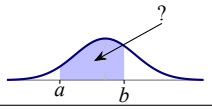
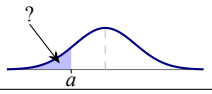
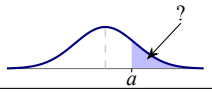
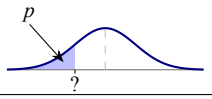
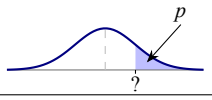
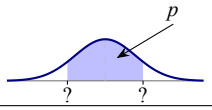
| Item | Result | Note |
|---|---|---|
| Standardise | $Z = \dfrac{X-\mu}{\sigma} \sim \mathrm{N}(0, 1)$ | $Z$ = number of s.d. from the mean |
| Symmetry | $\mathrm{P}(X < \mu - a) = \mathrm{P}(X > \mu + a)$ | $\mathrm{P}(Z < -z) = \mathrm{P}(Z > z) = 1 - \mathrm{P}(Z < z)$ |
| Complement | $\mathrm{P}(X > a) = 1 - \mathrm{P}(X < a)$ | total area is 1 |




Left: equal tails. Right: $\mathrm{P}(\mu - a < X < \mu + a) = p$, so each tail is $\frac{1-p}{2}$.

## 5.2 Finding Probabilities and Values (GC)

| Find | GC command | Note | Graph |
|---|---|---|---|
| $P(a<X<b)$ | `normalcdf(a,b,μ,σ)` | enter $\sigma$, **not** $\sigma^2$ |   |
| $P(X<a)$ | `normalcdf(-1E99,a,μ,σ)` | 1 2nd , 9 9 for 1E99 |   |
| $P(X>a)$ | `normalcdf(a,1E99,μ,σ)` | |   |
| $x$ with $P(X<x)=p$ | `invNorm(p,μ,σ,LEFT)` | **invNorm** gives the **left** area |   |
| $x$ with $P(X>x)=p$ | `invNorm(p,μ,σ,RIGHT)` | |   |
| $x$ with $P(\mu-c<X<\mu+c)=p$ | `invNorm(p,μ,σ,CENTER)` | |   |

**GC:** 2nd VARS (DISTR) → `normalcdf(`, `invNorm(`. If $\mu, \sigma$ are omitted, the GC uses N(0, 1).

**Example 5.1**

$X \sim N(60, 8^2)$: find $P(52<X<70)$, and $x$ with $P(X>x)=0.2$

$$P(52<X<70)=0.736$$

$$P(X<x)=0.8$$

$$x=\text{invNorm}(0.8, 60, 8)$$

$$=66.7$$

**Example 5.2**

Find $\mu$, $\sigma$ given $P(X<30)=0.1$ and $P(X>46)=0.2$

$$\frac{30-\mu}{\sigma}=\text{invNorm}(0.1)=-1.2816$$

$$\frac{46-\mu}{\sigma}=\text{invNorm}(0.8)=0.8416$$

$$\frac{16}{\sigma}=2.1232$$

$$\sigma=7.54$$

$$\mu=39.7$$

## 5.3 Linear Functions and Combinations of Normals

| Situation | Distribution | Condition |
|---|---|---|
| $aX+b$ | $\mathrm{N}(a\mu+b, a^2\sigma^2)$ | $X$ normal |
| $aX\pm bY$ | $\mathrm{N}(a\mu_X\pm b\mu_Y, a^2\sigma_X^2+b^2\sigma_Y^2)$ | $X$, $Y$ independent normals |
| $X_1+X_2+\cdots+X_n$ | $\mathrm{N}(n\mu, n\sigma^2)$ | independent copies of $X$ |
| $nX$ | $\mathrm{N}(n\mu, n^2\sigma^2)$ | one observation $\times n$ |

**Common mistake**

× Variances **add** for both $X+Y$ and $X-Y$: $\mathrm{Var}(X-Y)=\mathrm{Var}(X)+\mathrm{Var}(Y)$. $\mathrm{Var}(2X)=4\mathrm{Var}(X)\neq\mathrm{Var}(X_1+X_2)=2\mathrm{Var}(X)$. A comparison such as $X>Y$ is rewritten as $X-Y>0$, then one normal variable.

**Example 5.3**

$X\sim\mathrm{N}(120, 15^2)$, $Y\sim\mathrm{N}(100, 8^2)$ independent; $\mathrm{P}(X>Y)$

$$X-Y\sim\mathrm{N}(20, 15^2+8^2)$$
$$=\mathrm{N}(20, 17^2)$$
$$\mathrm{P}(X>Y)=\mathrm{P}(X-Y>0)$$
$$=0.880$$

## 5.4 Further Question Types

| Type | Method |
|---|---|
| Percentage change ("loses 5%") | new variable is $0.95X$: $\mathrm{N}(0.95\mu, 0.95^2\sigma^2)$ (not $X-5$) |
| Total of different items (e.g. 2 of $X$ and 3 of $Y$) | $T=X_1+X_2+Y_1+Y_2+Y_3$: $\mathrm{E}(T)=2\mu_X+3\mu_Y$, $\mathrm{Var}(T)=2\sigma_X^2+3\sigma_Y^2$; with scaled items add $a^2\sigma^2$ |
| "differs by less than $c$" | $\mathrm{P}(\lvert X-Y\rvert<c)=\mathrm{P}(-c<X-Y<c)$: `normalcdf(-c,c,μ,σ)` |
| Unknown $\mu$ or $\sigma$ (one condition) | standardise: $\frac{a-\mu}{\sigma}=$ `invNorm(p)`, solve (the sign of $z$ follows the tail) |
| Range of $\sigma$ (or $\mu$) for $\mathrm{P}(X>a)<p$ | $\frac{a-\mu}{\sigma}>$ `invNorm(1-p)`; $\sigma>0$, so multiply through without reversing; then solve |
| Normal event, then count ("$k$ of $n$ items") | $p=\mathrm{P}$(event) by GC; then $\mathrm{B}(n,p)$ |
| Sketch of normal curves | bell shape, axis marked at $\mu$; same $\mu$: larger $\sigma$ is wider and lower; different $\mu$: curves shifted; each has area 1 |
| Model unsuitable | compute $\mathrm{P}(X<0)$ (or outside a natural range): not negligible; or non-symmetric data |

Where asked to "state the parameters": write each distribution with numbers, e.g. $T\sim\mathrm{N}(190, 72)$. Assume independence (e.g. masses of different items do not affect one another). Normal approximation to a binomial distribution is not in the H2 syllabus.

**Example 5.4**

$X \sim \mathrm{N}(50, 3^2)$, $Y \sim \mathrm{N}(30, 2^2)$ independent; $T = 2X+3Y$, find $\mathrm{P}(T>200)$

$$\mathrm{E}(T) = 2(50)+3(30) = 190$$

$$\mathrm{Var}(T) = 4(9)+9(4) = 72$$

$$T \sim \mathrm{N}(190, 72)$$

$$\mathrm{P}(T>200) = 0.119$$

**Example 5.5**

$X \sim \mathrm{N}(60, \sigma^2)$, $\mathrm{P}(X>70) < 0.1$

$$\frac{70-60}{\sigma} > \mathrm{invNorm}(0.9) = 1.2816$$

$$\sigma < \frac{10}{1.2816}$$

$$\therefore 0 < \sigma < 7.80$$

# 6 Sampling and the Central Limit Theorem

## 6.1 Population and Sample

| Term | Meaning | Notation |
|---|---|---|
| Population | the whole group of interest | mean $\mu$, variance $\sigma^2$ (parameters) |
| Sample | a part of the population, size $n$ | $\bar{x}$, $s^2$ (statistics) |
| Census | every member is surveyed: the replies are the **population**; a part of them is a **sample** | |
| Simple random sample | every member (and every sample of size $n$) is equally likely to be chosen; members chosen independently | |
| To obtain one | list and number the whole population; choose numbers by a random number generator; ignore repeats | |

| Not random if … | Reason |
|---|---|
| chosen by convenience (e.g. first 30 in the canteen, volunteers) | some members are more likely than others |
| only part of the population can be chosen | the rest has no chance |
| one choice affects another (e.g. friends chosen together) | not independent |

Advantage of a random sample: it avoids bias, since every member has the same chance and members are chosen independently.

## 6.2 Unbiased Estimates

| Estimate of | Raw data | Coded: $\Sigma(x-a)$, $\Sigma(x-a)^2$ |
|---|---|---|
| Mean $\mu$ | $\bar{x}=\dfrac{\sum x}{n}$ | $\bar{x}=a+\dfrac{\sum(x-a)}{n}$ |
| Variance $\sigma^2$ | $s^2=\dfrac{1}{n-1}\left[\sum x^2-\dfrac{\left(\sum x\right)^2}{n}\right]$ | $s^2=\dfrac{1}{n-1}\left[\sum(x-a)^2-\dfrac{\left(\sum(x-a)\right)^2}{n}\right]$ |
| Also | $s^2=\dfrac{1}{n-1}\sum(x-\bar{x})^2$ | MF27 |

**GC:** STAT → CALC → 1-Var Stats. Use Sx (= $s$, divides by $n-1$), so $s^2=$ Sx$^2$. For a frequency table, enter the values in L1 and the frequencies in L2, then 1-Var Stats L1,L2. Do **not** use $\sigma$x (divides by $n$; this is not the unbiased estimate).

**Example 6.1**

$n=10, \sum x=125, \sum x^2=1830$

$$\bar{x}=\frac{125}{10}=12.5$$

$$s^2=\frac{1}{9}\left[1830-\frac{125^2}{10}\right]$$

$$=\frac{267.5}{9}=29.7$$

**Example 6.2**

given $\sum(x-10)=25, \sum(x-10)^2=330$

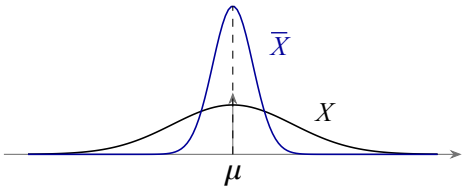
$$\bar{x}=10+\frac{25}{10}=12.5$$

$$s^2=\frac{1}{9}\left[330-\frac{25^2}{10}\right]$$

$$=\frac{267.5}{9}=29.7$$

## 6.3 Distribution of the Sample Mean

| Population | Distribution of $\bar{X}$ | Reason |
|---|---|---|
| Normal, $X\sim \mathrm{N}(\mu,\sigma^2)$ | $\bar{X}\sim \mathrm{N}(\mu,\frac{\sigma^2}{n})$ (exact) | any $n$; linear combination of normals |
| Not normal / unknown, $n$ **large** (e.g. $n\geqslant 30$) | $\bar{X}\approx \mathrm{N}(\mu,\frac{\sigma^2}{n})$ (approx.) | **Central Limit Theorem** |
| Any population | $\mathrm{E}(\bar{X})=\mu,\ \mathrm{Var}(\bar{X})=\frac{\sigma^2}{n}$ | true for every $n$ |




Same centre $\mu$. $\bar{X}$ is **narrower**: s.d. $\frac{\sigma}{\sqrt{n}}$ (larger $n$, narrower).

**Example 6.3**

population mean 80, s.d. 12, $n=40$

$$\bar{X}\approx \mathrm{N}(80,\frac{12^2}{40}) \text{ (CLT)}$$

$$\mathrm{P}(\bar{X}>83)=0.0569$$

**Central Limit Theorem**

For a random sample of size $n$ (large, e.g. $n\geqslant 30$) from **any** population with mean $\mu$ and variance $\sigma^2$: $\bar{X}\approx \mathrm{N}(\mu,\frac{\sigma^2}{n})$

Likewise $\sum_{i=1}^{n} X_i \approx \mathrm{N}(n\mu, n\sigma^2)$. If $n$ is small, the population must be normal.

## 6.4 Which Distribution?

| The question … | Use | Key step |
|---|---|---|
| number of successes in $n$ independent trials | $\mathrm{B}(n,p)$ | state the conditions |
| one measurement, normal | $\mathrm{N}(\mu,\sigma^2)$ | standardise / GC |
| total or difference of independent normals | $\mathrm{N}(\cdot,\cdot)$ | add the **variances** |
| mean of $n$ values, population normal | $\bar{X}\sim \mathrm{N}(\mu,\sigma^2/n)$ | exact |
| mean of $n$ values, population not normal, $n$ large | $\bar{X}\approx \mathrm{N}(\mu,\sigma^2/n)$ | say "by CLT" |

# 6.5 Further Question Types

| Type | Method |
|---|---|
| Population given by a table or $B(n,p)$, $n$ observations large | $\mu = E(X)$, $\sigma^2 = \text{Var}(X)$ first (Ch 3, 4); then $\overline{X} \approx N(\mu, \sigma^2/n)$ by CLT |
| Total of many independent counts (e.g. defects in 100 batches) | $S = X_1 + \cdots + X_n \approx N(n\mu, n\sigma^2)$ by CLT; for different counts add the means and the variances |
| Least $n$ for $P(\overline{X} < c) \geqslant p$ | $\frac{c-\mu}{\sigma/\sqrt{n}} \geqslant$ **invNorm(p)**, solve for $\sqrt{n}$; or GC table: $Y_1 =$ **normalcdf** with s.d. $\sigma/\sqrt{X}$, X as $n$ |
| Is the sample size big enough? | $n < 30$ and population not known to be normal: CLT **cannot** be used; $n \geqslant 30$: say "by CLT" |

**Example 6.4**

$X \sim B(12, 0.25)$, $n = 40$; $P(\overline{X} > 3.4)$

$$\mu = 12(0.25) = 3$$

$$\sigma^2 = 12(0.25)(0.75) = 2.25$$

$$\overline{X} \approx N\left(3, \frac{2.25}{40}\right) \text{ (CLT)}$$

$$P(\overline{X} > 3.4) = 0.0458$$

**Example 6.5**

same $X$; least $n$ with $P(\overline{X} < 3.3) \geqslant 0.9$

$$\frac{3.3-3}{1.5/\sqrt{n}} \geqslant \text{invNorm}(0.9) = 1.2816$$

$$\sqrt{n} \geqslant 6.408$$

$$n \geqslant 41.1$$

$$\therefore n_{\min} = 42$$

# 7 Hypothesis Testing

## 7.1 Terminology

| Term | Meaning |
|---|---|
| Null hypothesis $H_0$ | the claim assumed true: $H_0: \mu = \mu_0$ |
| Alternative hypothesis $H_1$ | what we look for evidence of: $\mu < \mu_0$, $\mu > \mu_0$ (1-tail) or $\mu \neq \mu_0$ (2-tail) |
| Test statistic | $\bar{X}$ (or $Z$), whose distribution is found assuming $H_0$ is true |
| Level of significance $\alpha$ | probability of rejecting $H_0$ when $H_0$ is true; given in the question |
| Critical region | values of the test statistic for which $H_0$ is rejected; boundary = critical value |
| $p$-value | probability, assuming $H_0$ true, of a test statistic at least as extreme as the one observed; smallest level at which $H_0$ is rejected |

## 7.2 Formulating the Hypotheses

| Wording | $H_1$ | Type |
|---|---|---|
| decreased, less than, lower, overstated | $\mu < \mu_0$ | 1-tail (lower) |
| increased, more than, greater, understated | $\mu > \mu_0$ | 1-tail (upper) |
| changed, different, not correct, biased | $\mu \neq \mu_0$ | 2-tail |

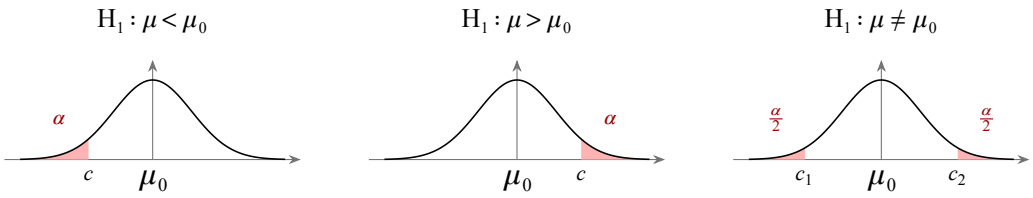
Define $\mu$ in context (e.g. "$\mu$ is the population mean mass"). $H_0$ always uses "=".

## 7.3 Test Statistic and Its Distribution

| Situation | Distribution under $H_0$ | Say |
|---|---|---|
| Normal population, $\sigma^2$ known | $\bar{X} \sim N(\mu_0, \frac{\sigma^2}{n})$ | any $n$ |
| Any population, $n$ large, $\sigma^2$ known | $\bar{X} \approx N(\mu_0, \frac{\sigma^2}{n})$ | "by CLT" |
| Any population, $n$ large, $\sigma^2$ unknown | $\bar{X} \approx N(\mu_0, \frac{s^2}{n})$ | "by CLT"; $s^2$ unbiased estimate |

Random sample assumed. Small $n$ with unknown $\sigma^2$ is not tested in H2 (no $t$-test). Small $n$ needs "population is normal" to be given or assumed.

## 7.4 The Test: *p*-value and Critical Region




| | **Lower tail** | **Upper tail** | **2-tail** |
|---|---|---|---|
| *p*-value | $P(\bar{X} \leqslant \bar{x})$ | $P(\bar{X} \geqslant \bar{x})$ | $2P(\bar{X} \leqslant \bar{x})$ if $\bar{x} < \mu_0$;<br>$2P(\bar{X} \geqslant \bar{x})$ if $\bar{x} > \mu_0$ |
| Reject $H_0$ if | *p*-value $\leqslant \alpha$ | | |
| Critical region | $\bar{X} \leqslant c$, $P(\bar{X} \leqslant c) = \alpha$ | $\bar{X} \geqslant c$, $P(\bar{X} \geqslant c) = \alpha$ | $\bar{X} \leqslant c_1$ or $\bar{X} \geqslant c_2$; each tail $\frac{\alpha}{2}$ |
| Reject $H_0$ if | $\bar{x}$ lies in the critical region | | |

| **Critical** $z$ | $\alpha = 5\%$ | $\alpha = 1\%$ | $\alpha = 10\%$ | $\alpha = 2\%$ |
|---|---|---|---|---|
| 1-tail (upper) | 1.645 | 2.326 | 1.282 | 2.054 |
| 2-tail | $\pm 1.960$ | $\pm 2.576$ | $\pm 1.645$ | $\pm 2.326$ |

Using $z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$: reject $H_0$ if $z$ is in the critical region (lower tail: $z \leqslant -1.645$). The same decision as the *p*-value test.

**Steps of a test**

1 $H_0: \mu = \mu_0$, $H_1: \mu \square \mu_0$; define $\mu$; state $\alpha$
2 distribution of $\bar{X}$ under $H_0$ (with reason: normal population / CLT)
3 *p*-value from the GC (or critical region)
4 compare: $p \leqslant \alpha \Rightarrow$ reject $H_0$; $p > \alpha \Rightarrow$ do not reject $H_0$
5 conclusion **in context**

| **Result** | **Conclusion (in context)** |
|---|---|
| Reject $H_0$ | there is sufficient evidence, at the $\alpha$ level of significance, to conclude that [$H_1$ in words] |
| Do not reject $H_0$ | there is insufficient evidence, at the $\alpha$ level of significance, to conclude that [$H_1$ in words] |

Never write "accept $H_0$" or "$H_0$ is true / proven".

**GC:** STAT → TESTS → 1:Z-Test. Inpt:Stats; enter $\mu 0$, $\sigma$ (use $s$ if $\sigma$ unknown, large $n$), $\bar{x}$, n; choose $\neq \mu 0$, $< \mu 0$ or $> \mu 0$ to match $H_1$; Calculate gives $z$ and $p$. With raw data use Inpt:Data with the list; for a frequency table first run 1-Var Stats L1,L2 to get $\bar{x}$ and Sx, then use Inpt:Stats.

**Example 7.1**

$H_0: \mu = 500$, $H_1: \mu < 500$; $\sigma = 10$, $n = 50$, $\bar{x} = 497$, $\alpha = 5\%$

$$\overline{X} \sim N(500, \frac{10^2}{50}) \text{ under } H_0 \text{ (CLT)}$$

$$p\text{-value} = P(\overline{X} \leqslant 497) = 0.0169$$

$$\because 0.0169 \leqslant 0.05$$

$$\therefore \text{reject } H_0$$

**Example 7.2**

$H_0: \mu = 25$, $H_1: \mu \neq 25$; $n = 36$, $\bar{x} = 26.2$, $s^2 = 9$

$$\overline{X} \approx N(25, \frac{9}{36}) \text{ under } H_0 \text{ (CLT)}$$

$$z = \frac{26.2 - 25}{0.5} = 2.4$$

$$p\text{-value} = 2P(\overline{X} \geqslant 26.2) = 0.0164$$

$0.0164 > 0.01 \Rightarrow$ do not reject $H_0$ ($\alpha = 1\%$)

$0.0164 \leqslant 0.05 \Rightarrow$ reject $H_0$ ($\alpha = 5\%$)

## 7.5 Using the Result of a Test to Find Unknowns

| Given | Method |
|---|---|
| $H_0$ rejected at level $\alpha$; find range of $\alpha$ | $\alpha \geqslant p$-value (smallest level of significance $= p$-value) |
| Range of $\bar{x}$ (or $\mu_0$, $n$) for which $H_0$ is rejected | critical region: find $c$ by **invNorm**, then write the inequality in $\bar{x}$ |
| Range of $n$ | $p$-value (or $z$) in terms of $n$; use a GC table in $n$ to find where the condition holds |
| $p$-value given; find $\sigma$ (or $\bar{x}$, $n$) | one-tail: $\frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} =$ **invNorm(1-p)** (upper) or **invNorm(p)** (lower); two-tail: use $\frac{p}{2}$ in the tail |
| One more data value $h$ added to the sample | new mean $= \frac{n\bar{x} + h}{n+1}$; apply the critical region (or $p$-value condition) and solve the inequality for $h$ |
| Values of $\bar{x}$ for which $H_0$ is **not** rejected | the region **between** the critical values (2-tail), or on the other side of $c$ (1-tail) |
| Effect of changing $\sigma$, $n$ or $\alpha$ | larger $\sigma$ (or smaller $n$): $\sigma/\sqrt{n}$ larger, $p$-value larger, harder to reject; smaller $\sigma$ (or larger $n$): easier to reject; larger $\alpha$: easier to reject |
| Assumption needed (small $n$) | population normal; and $\sigma^2$ unchanged / known (state which). Large $n$: none, by CLT |
| Sample not suitable for the test | sample not random (bias), or $n$ too small for CLT when the population is not known to be normal |
| One-tail or two-tail? | decided by the wording **before** looking at the data: a direction ("reduced", "higher") $\Rightarrow$ 1-tail; "changed" / "correct?" $\Rightarrow$ 2-tail |

**Example 7.3**

$H_1: \mu > 50$; $\sigma = 8$, $n = 40$, $\alpha = 5\%$. Critical region?

$$\overline{X} \sim N(50, \frac{8^2}{40})$$

$$c = \text{invNorm}(0.95, 50, \sqrt{1.6})$$

$$= 52.08$$

$$\therefore \text{reject if } \bar{x} \geqslant 52.08$$

**Example 7.4**

$H_0: \mu = 20$, $H_1: \mu > 20$; $n = 25$, $\bar{x} = 21.0$; $p$-value $= 0.0228$. Find $\sigma$

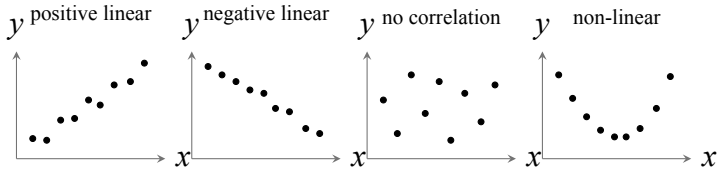
$$\frac{21.0 - 20}{\sigma/\sqrt{25}} = \text{invNorm}(1 - 0.0228)$$

$$\frac{5}{\sigma} = 2.00$$

$$\therefore \sigma = 2.50$$

# 8 Correlation and Linear Regression

## 8.1 Scatter Diagrams




| Item | Note |
|---|---|
| Independent $x$, dependent $y$ | $x$: controlled / explains; $y$: the response (e.g. $y$ = sales, $x$ = advertising) |
| Describe a scatter diagram | direction (positive / negative), linear or curved, strength (close to a line), outliers |
| Outlier | point far from the pattern; check data, comment on its effect on $r$ and on the line |
| GC | enter $x$ in **L1**, $y$ in **L2**; 2nd Y= (STAT PLOT) → **Plot1** on, scatter; ZOOM **9:ZoomStat** |

## 8.2 Product Moment Correlation Coefficient $r$

**PMCC (MF27)**

$$r=\frac{\sum(x-\bar{x})(y-\bar{y})}{\sqrt{\sum(x-\bar{x})^2\sum(y-\bar{y})^2}}=\frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}$$

$$S_{xx}=\sum x^2-\frac{\left(\sum x\right)^2}{n},\quad S_{yy}=\sum y^2-\frac{\left(\sum y\right)^2}{n},\quad S_{xy}=\sum xy-\frac{\sum x\sum y}{n}$$

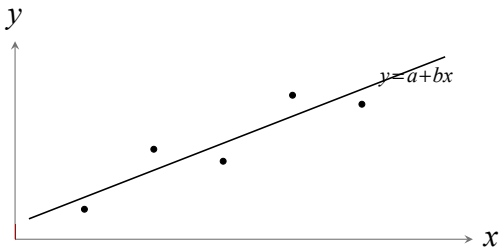
| Value of $r$ | Interpretation |
|---|---|
| $r=1$ or $-1$ | all points lie exactly on a straight line (positive / negative gradient) |
| $r$ close to 1 | strong positive linear correlation |
| $r$ close to $-1$ | strong negative linear correlation |
| $r$ close to 0 | little or no **linear** correlation (a non-linear relation may still exist) |

Note

$-1\leqslant r\leqslant 1$. $|r|$ close to 1 only says a **linear** model fits well; check the scatter diagram. A strong correlation does **not** mean that $x$ causes $y$.

**GC:** 2nd 0 (CATALOG) → **DiagnosticOn** (once). Then STAT → **CALC** → **4:LinReg(ax+b) L1,L2** gives **a**, **b**, **r**. Use **8:LinReg(a+bx)** for $y=a+bx$.

## 8.3 Least-Squares Regression Line




Line of $y$ on $x$ minimises $\sum$(vertical distances)$^2$ (red).

| | $y$ **on** $x$ | $x$ **on** $y$ |
|---|---|---|
| Equation | $y-\bar{y}=b(x-\bar{x}),\ b=\frac{S_{xy}}{S_{xx}}$ | $x-\bar{x}=d(y-\bar{y}),\ d=\frac{S_{xy}}{S_{yy}}$ |
| Or | $y=a+bx,\ a=\bar{y}-b\bar{x}$ | $x=c+dy,\ c=\bar{x}-d\bar{y}$ |
| Use to estimate | $y$ from a given $x$ | $x$ from a given $y$ |

**Note**

Both lines pass through $(\bar{x},\bar{y})$. To estimate $x$ from $y$, use the line of $x$ on $y$ (do **not** rearrange $y$ on $x$), whichever variable is the independent one. The gradient $b$ has the sign of $r$.

**Example 8.1**

$x$: 1, 2, 3, 4, 5, 6 $y$: 3.1, 4.3, 5.2, 6.8, 7.1, 8.9

$$\sum x=21,\ \sum y=35.4,\ \sum xy=143.4,\ \sum x^2=91,\ \sum y^2=231$$

$$S_{xx}=91-\frac{21^2}{6}=17.5$$

$$S_{xy}=143.4-\frac{21(35.4)}{6}=19.5$$

$$S_{yy}=231-\frac{35.4^2}{6}=22.14$$

$$r=\frac{19.5}{\sqrt{17.5\times 22.14}}=0.991$$

$$b=\frac{19.5}{17.5}=1.114,\quad a=5.9-1.114(3.5)=2.00$$

$$\therefore y=2.00+1.11x$$

## 8.4 Interpolation, Extrapolation and Interpretation

| Item | Note |
|---|---|
| Interpolation | estimate for $x$ **within** the data range: reliable if $\|r\|$ is close to 1 and the scatter diagram is linear |
| Extrapolation | estimate for $x$ **outside** the data range: unreliable, the linear relation may not continue |
| Gradient $b$ | the change in $y$ per unit increase in $x$ (in context) |
| Intercept $a$ | value of $y$ when $x=0$; meaningful only if $x=0$ is sensible in context |
| How well does the model fit? | $\|r\|$ close to 1 and points close to the line ⇒ good fit; comment on the range of $x$ |

**Example 8.2**

using the example above: $x=3.5$ and $x=10$

$$x=3.5: \ y=2.00+1.114(3.5)=5.90 \ \text{(interpolation, reliable)}$$

$$x=10: \ y=2.00+1.114(10)=13.1 \ \text{(extrapolation, unreliable)}$$

## 8.5 Transformations to Achieve Linearity

Choose the model whose scatter diagram of (transformed variable, $y$) is the most linear, and whose $|r|$ is closest to 1.

| Model | Scatter diagram | Plot | Regression | Shape |
|---|---|---|---|---|
| $y=a+bx^2$ |  | $y$ against $x^2$ | $y$ on $x^2$ | increasing, curving up |
| $y=a+b\ln x$ |  | $y$ against $\ln x$ | $y$ on $\ln x$ | increasing, curving down |
| $y=a+\frac{b}{x}$ |  | $y$ against $\frac{1}{x}$ | $y$ on $\frac{1}{x}$ | decreasing, levelling off |
| $y=a\,e^{bx}$ |  | $\ln y$ against $x$ | $\ln y$ on $x$ | increasing, curving up faster |

Let $w$ be the transformed variable (e.g. $w=x^2$, $\ln x$, $\frac{1}{x}$). Find $r$ and the line of $y$ on $w$ with the GC (new list), then substitute $w=\ldots$ to estimate $y$. For $\ln y$ on $x$: $\ln y=a+bx \Rightarrow y=e^a e^{bx}$.

## 8.6 Further Question Types

| Type | Method |
|---|---|
| Residual and least squares | residual $e = y - (a+bx)$ (observed − predicted); positive above the line. For the least-squares line $\sum e = 0$, so **squares** are used; the line minimising $\sum e^2$ is the best fit; smaller $\sum e^2 \Rightarrow$ better fit |
| Sum of squares of residuals | compute each $e$, square, add (or use the GC); compare two lines or two models by their $\sum e^2$ |
| Missing data value / unknown | the line passes through $(\bar{x}, \bar{y})$: $\bar{y} = a + b\bar{x}$ with the unknown in $\sum x$ or $\sum y$; solve |
| Extra point that leaves the line unchanged | the point $(\bar{x}, \bar{y})$ (the mean point) |
| Change of units or coding, $x' = px+q$ | $r$ unchanged if $p>0$ (sign of $r$ reverses if $p<0$); new line: substitute $x = \frac{x'-q}{p}$ into the fitted equation |
| Remove or change one point | recompute $r$, $a$, $b$ on the GC; comment on the effect (e.g. $\lvert r \rvert$ nearer 1) |
| Choosing a model | scatter shape **in context** (e.g. levels off, cannot be negative), then $\lvert r \rvert$ closest to 1; state the constants and $r$ for the chosen model |
| Estimate $x$ from $y$ (transformed model) | solve the fitted equation for $x$ (e.g. $y = a + \frac{b}{x} \Rightarrow x = \frac{b}{y-a}$), or GC graph intersect |
| Sketch a scatter diagram | label axes with scales; plot every point (label any special point); do not join; show the shape (curve, outlier) |

**GC:** `LinReg(a+bx) L1,L2,Y1` ([VARS] → Y-VARS → `1:Function` → `Y1` stores the line). In `L3` enter `L2-Y1(L1)`. Then [2nd] [STAT] (LIST) → MATH → `5:sum( L3²)` gives $\sum e^2$. Transformed variable: highlight the `L3` header and enter e.g. `ln(L1)`, `L1²` or `1/L1`, then run `LinReg` with `L3,L2`.

**Example 8.3**

$x$: 1, 2, 3, 4 $y$: 3, 6, 6, 10; line $y = 2x+1$

$$e = 0,\ 1,\ -1,\ 1$$

$$\sum e^2 = 0+1+1+1 = 3$$

least squares $y = 1 + 2.1x$

$$\sum e^2 = 2.70 < 3$$

**Example 8.4**

$s = 1.2t + 5$, $t$ in °C; use $F$ in °F

$$t = \frac{5}{9}(F-32)$$

$$s = 1.2 \times \frac{5}{9}(F-32) + 5$$

$$= \frac{2}{3}F - 16.3$$